Project Topic 10: Analyze Operational Data to Improve Logistics & Warehousing Efficiency (*LogiTech*)

Context:

A leading logistics provider has curated a comprehensive dataset to analyze shipment performance, warehouse operations, and inventory movement across multiple regions. The dataset integrates shipment delivery logs, warehouse operations metrics, and inventory stock movement records. As a data analyst, your role is to explore and analyze this dataset to help the organization answer critical aspects such as efficiency of deliveries, warehouse utilization, inventory turnover, and operational costs.

Business Objective 1: To improve delivery reliability and cost efficiency by analyzing delays, accuracy, and regional performance.

Sample Business questions

- Which routes or regions have the highest delivery delays?
- What is the average delivery time and shipment cost per region?

Business Objective 2: To optimize warehouse capacity, labor productivity, and fulfilment speed in order to enhance operational efficiency.

Sample Business questions

- Which warehouses operate near or above capacity?
- How does warehouse utilization vary with labour hours and operational costs?

Business Objective 3: To strengthen inventory management by minimizing stockouts, balancing demand, and reducing carrying costs across product categories.

Sample Business questions

- Which product categories show frequent stockouts or under-stocking?
- How does inventory turnover vary by warehouse or product category?

Business Objective 4: To maximize financial efficiency by controlling operational costs, optimizing procurement, and aligning budgets with critical product flows.

Sample Business questions

- What is the operational cost per day per warehouse, and how can it be optimized?
- How does cost per unit variation affect warehouse efficiency and budget allocation?